Engineering Mathematics - I · Unit_03 · Unit 3

DRAFT — UE26MA141A · Unit_03 · Unit 3 · Solution of one dimensional heat equation

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Solution of one dimensional heat equation

1 Solution of one dimensional heat equation

Study focus: Solution of one dimensional heat equation. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000102 · EV0000103

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

In previous classes, we understood what is the method of separation of variables.

Key details

- Why did we use method of separation of variables?
- We also solved some of the problems using method of separation of variables.
- In this session also, we will find the various possible solutions of one dimensional heat equation by using method of separation of variables.
- It means x and t are independent variables.
- In the form of a product that is u equal to x t where x is a function of x only and t is a function of t only.
- Then C square the right hand side term can be written as t of d square x divided by dx square.
- If you look at the LH system, we need to remove this X.
- If you look at the right hand side, we need we need to remove this T.
- C square by capital X, T square X divided by P x square.
- This can also be written as 1 by C square t dt by dt equals 1 over x d square x divided by dx square.
- We have expressed the left hand side as a function of P and the right hand side as a function of X.
- And similarly 1 by c square t.
- D by D T equals.
- of x equals zero where capital D stands for D by DX.
- In a similar manner.

Example or application from the lecture

- As in Open Theory, the one the dimensional heat equation is given by dou u by dou t equals c square times dou square u divided by dou x square.
- Then we need to substitute this assumption into the given.
- After we substitute u equals x t into the given problem, we need to convert the partial derivatives present in this equation as the ordinary derivatives so this is nothing but the left hand height left hand side can be written as x of dt by dt.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000102 · EV0000103 · EV0000104 · +1 source units in pack.json

Core explanation · 05:00–09:59

C square capital T when you rearrange this is written as T minus K C square of E equals zero.

Key details

- Mathematically, K is an arbitrary real constant, but different signs of k gives different ordinary differential equations in X and D.
- When you write axelor equation that gives m square equals zero.
- It means m equals zero comma zero two times.
- In a similar manner, this second expression becomes.
- This is M as D of P equals zero.
- What is the solution of these ordinary differential equations?
- If you observe X is a dependent and capital T is dependent, so we have to express capital X in terms of small X.
- As from second chapter, when the roots are repeated.
- Therefore, what is the general solution for the PD?
- In the later stage, we have to decide whether we accept this or not.
- We'll discuss this after the completion of the problem.
- That is when k equal to k n k is p square.
- then this differential equation becomes d square x divided by e x square minus p square x zero.
- X equals 0.
- This is same as t minus.
- P squared C square of P equals zero.
- Therefore accelerate equation becomes M equals sorry M minus P squared C square zero.

Example or application from the lecture

- Case number one, when k is zero, case number two, when k is positive, let us say p square, k is negative, let us say k equal to minus p square.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000106 · EV0000107 · EV0000108 · +1 source units in pack.json

Core explanation · 10:00–14:59

Again, how do we write the general solution when the roots are real or distinct?

Key details

- So for M equal to plus R minus P, the general solution will be okay now for convenient purpose, because we have used C1, C2, C3, we will now use C1 dash, C2 dash C3 dash.
- U equals xp.
- Then e power.
- P square C square T.
- Again, we need to discuss after the end of the problem.
- Should we accept this or should we reject this?
- When k is negative, say minus b square.
- As you can see, when we proceed based on case two, we get this.
- I mean to say when k is negative that is k is minus p square then the ordinary differential equations are going to be.
- You see the roots are complex.
- sorry the first root will be m plus p square c square zero so this is only m equals minus p square c square so this is the root.
- We write the general solution to be this as well as this.
- When K is zero, K is positive, and K is negative.
- And that's what we see in expression number 567.
- This is when a case positive C one dash, C two dash and C three dash.
- These are the various possible solutions of one dimensional heat equation by using method of separation of variables.

Example or application from the lecture

- Therefore, the solution of the PDE is given by.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000110 · EV0000111 · EV0000112 · +1 source units in pack.json

Core explanation · 15:00–19:59

Which solution is considered?

Key details

- In the sense Which solution is chosen, which solution is rejected.
- If you look at this seventh one, the seventh one is independent of the.
- In other words, since u equals c1x plus c2 times c3 is independent of t, this is rejected.
- It means expression number seven, expression seven is not considered.
- What about equation number five?
- Again you can see here as the time t increases this also increases it means as p increases u also increases therefore even the fifth one is also rejected.
- And observe E power minus C square P square t.

- It is easy to see that e power minus C square P square t decays to zero.
- As t tends to infinity.
- U decreases as the time t increases.
- That's what we see.
- Therefore, out of these three, the sixth one is considered.
- Of these three solutions, we have to choose that solution which is consistent with the physical nature of the problem.
- Since u decreases as time decreases, the only suitable solution of the 8th equation is the sixth one.
- So in this session we found out the various possible solutions of one dimensional heat equation by using the method of separation of variables.
- We also understood out of these three solutions, which one is suitable.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000114 · EV0000115 · EV0000116 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Solution of one dimensional heat equation transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/006_Solution_of_one_dimensional_heat_equation__77df3e42fb2d3295.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Solution of one dimensional heat equation transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/006_Solution_of_one_dimensional_heat_equation__77df3e42fb2d3295.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Solution of one dimensional heat equation transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/006_Solution_of_one_dimensional_heat_equation__77df3e42fb2d3295.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9985; only the JSON segmentation supplies the displayed timestamps.
Solution of one dimensional heat equation transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/006_Solution_of_one_dimensional_heat_equation__77df3e42fb2d3295.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
2. 1 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.
3. 1 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
4. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
5. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
6. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
7. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00034	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0035. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0036 to s0053. Sidecar variants compare...
SRC00035	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0035. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0036 to s0053. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00036	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0035. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0036 to s0053. Sidecar variants compare...
SRC00037	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0035. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0036 to s0053. Sidecar variants compare...

C Revision and retrieval practice